

PRACTICAL 13:

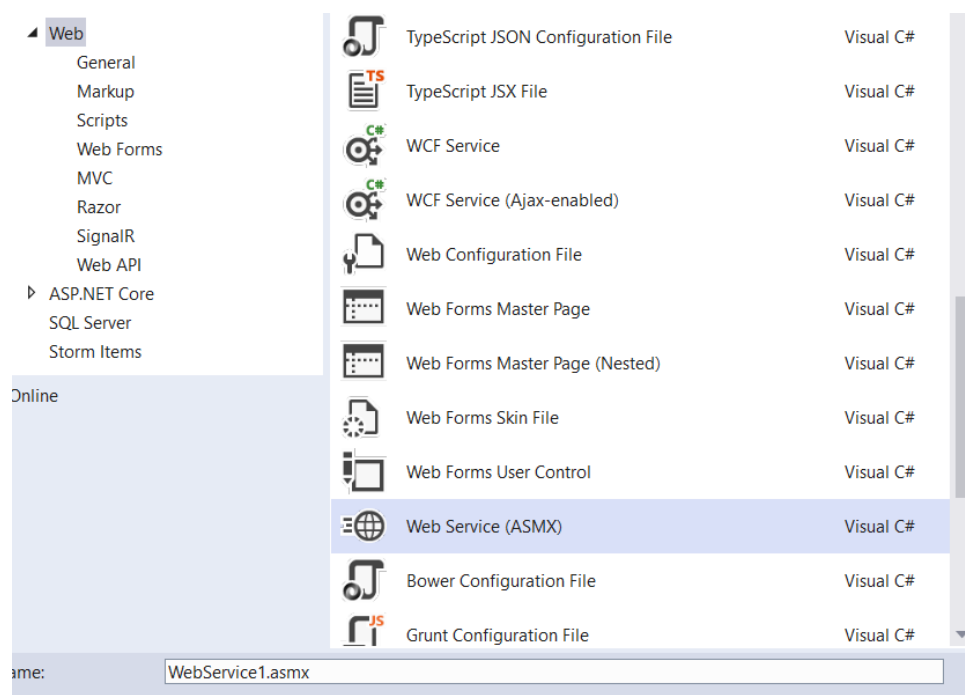
AIM: Design Web Application to Produce and Consume a Web Service.

STEP 1: CREATE PROJECT

1. Open Visual Studio
2. Click **Create New Project**
3. Select: ASP.NET Web Application (.NET Framework)
4. Select: Web Forms
5. Enter Project Name: **WebService**
6. Click Create

STEP 2: Add Web Service

- Right click project → **Add** → **New Item**
- Select **Web Service (ASMX)**



Step 3: Write Code:

Add methods inside class:

```
[WebMethod]
public string getname(string nm)
{
    return nm;
}

[WebMethod]
public int getaccountnumber(int a)
```

```
{
    return a;
}

// [System.Web.Script.Services.ScriptService]
0 references
public class WebService1 : System.Web.Services.WebService
{
    [WebMethod]
    0 references
    public string HelloWorld()
    {
        return "Hello World";
    }

    [WebMethod]
    0 references
    public string getname(string nm)
    {
        return nm;
    }

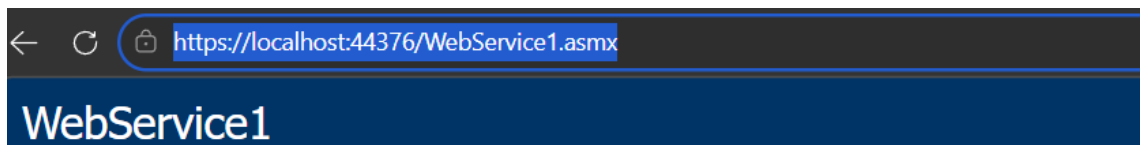
    [WebMethod]
    0 references
    public int getaccountnumber(int a)
    {
        return a;
    }
}
```

Step 4: Run Web Service

<https://localhost:xxxx/WebService1.asmx>

Ensure methods are visible:

- HelloWorld
- getname
- getaccountnumber



The following operations are supported. For a formal definition, please review the [Service Description](#).

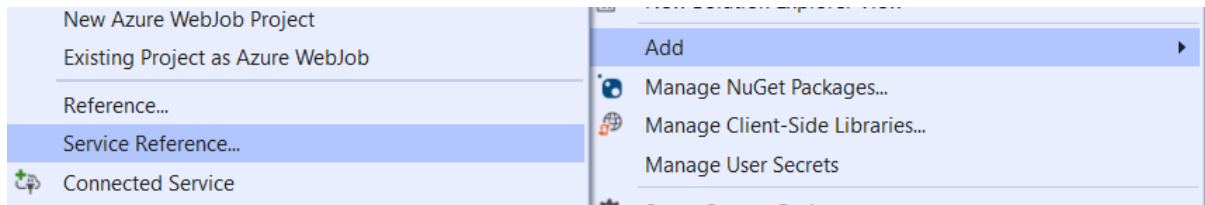
- [HelloWorld](#)
- [getaccountnumber](#)
- [getname](#)
- Copy URL

Step 5: Create Web Application(Comsumer)

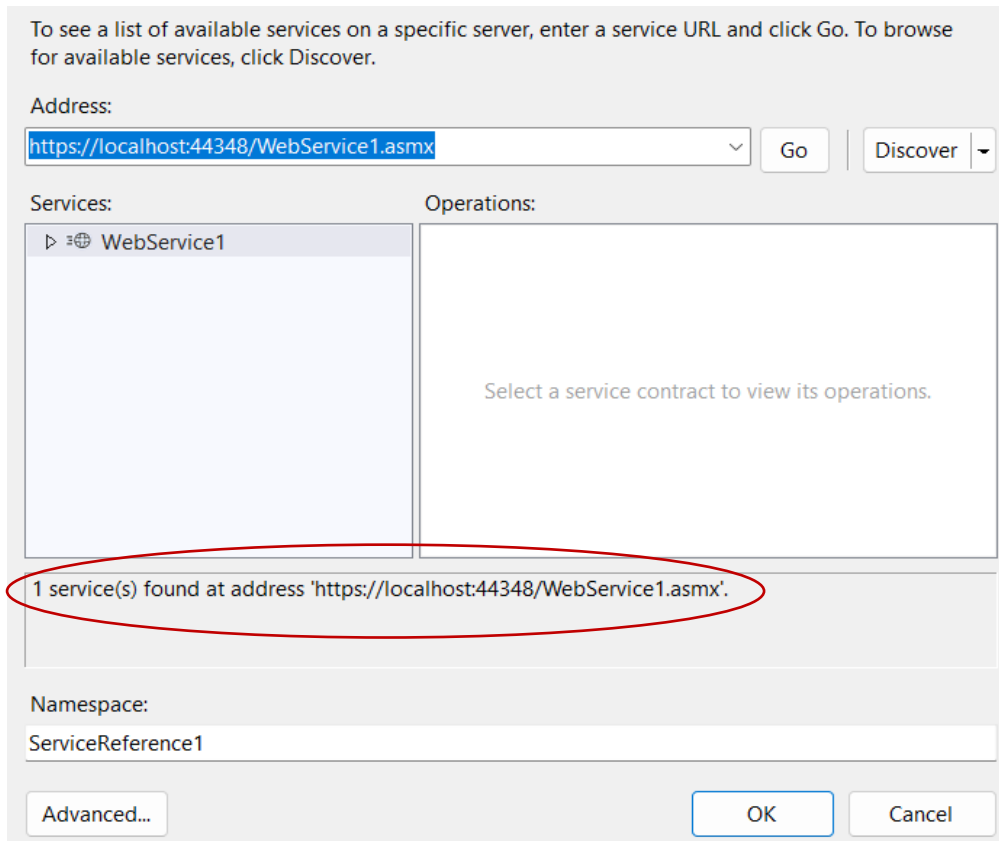
Add Web Form

Step 6: Add Web Reference

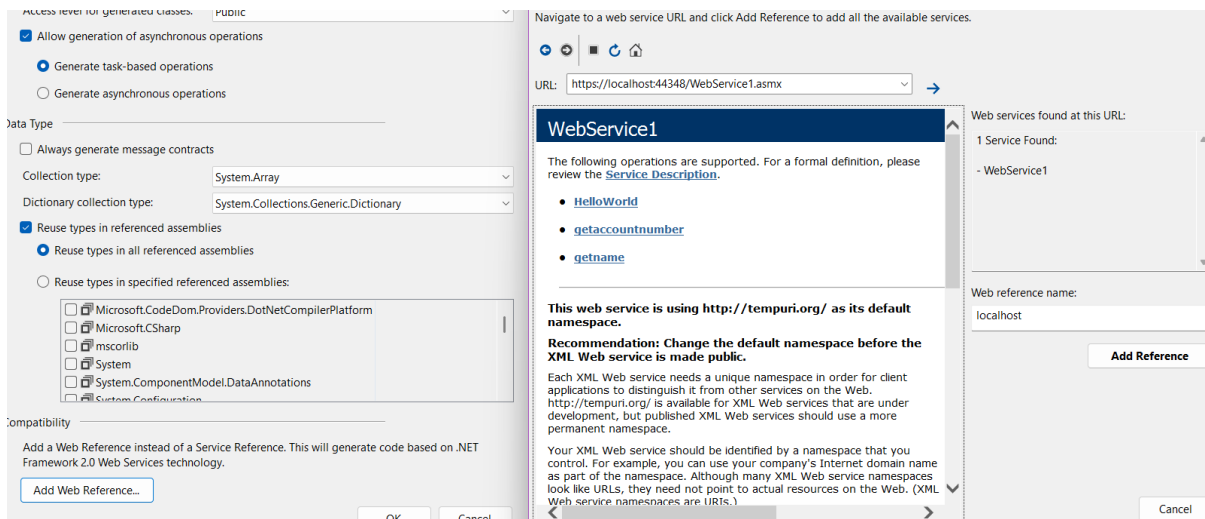
- Right click project → Add Service Reference



- Paste the URL
- Click Go



- Click Advanced
- Click Add Web Reference
- Paste the URL again



- Add Reference

Step 7: Design Web Page

Add:

- 2 TextBoxes
- 1 Button
- 2 Labels

WebForm1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebService.WebForm1" %>
```

```
<!DOCTYPE html>
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
```

```
<head runat="server">
```

```
<title></title>
```

```
</head>
```

```
<body>
```

```
<form id="form1" runat="server">
```

```
<div>
```

```
Enter Name:
```

```
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
```

```
<br /><br />
```

```
Enter Account No:
```

```
<asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
```

```
<br /><br />
```

```
<asp:Button ID="Button1" runat="server" Text="Show Data"
OnClick="Button1_Click" />
```

```
<br /><br />
```

```
<asp:Label ID="Label1" runat="server"></asp:Label>
```

```
<br />
```

```
<asp:Label ID="Label2" runat="server"></asp:Label>
```

```
</div>
```

```
</form>
```

```
</body>
</html>
```

Step 8: Write Code Behind

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace WebService
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Button1_Click(object sender, EventArgs e)
        {
            localhost.WebService1 obj = new localhost.WebService1();

            string name = obj.getname(TextBox1.Text);
            int acc = obj.getaccountnumber(Convert.ToInt32(TextBox2.Text));

            Label1.Text = "Name: " + name;
            Label2.Text = "Account No: " + acc;
        }
    }
}
```

Step 9: Run Web Form

- Right click WebForm1.aspx
- Set as Start Page and run.

Enter Name:

Enter Account No:

Name: pqrs

Account No: 498876